Syllabus Details

Syllabus ID:	14408
Syllabus Name:	**Digital Circuit Design_Thiết kế vi mạch số**
Course Name English:	**Digital Circuit Design**
Subject Code:	**DCD301**
Learning-Teaching Method:	Inquiry-Based Learning, In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	45h (60 sessions) contact hours + 1h final exam + 104h self-study
Pre-Requisite:	ECI101
Description:	This course provides students with a solid and comprehensive foundation in digital circuit design. It begins with essential concepts such as number systems, logic gates, Boolean algebra, and logic function simplification. Students then advance to designing and implementing combinational and sequential logic circuits, including multiplexers, arithmetic circuits, flip-flops, shift registers, counters, and memory elements. The course also introduces hardware description languages (VHDL/Verilog), FPGA-based implementation, and fundamental principles of digital signal and data processing. Through these topics, students learn how hardware is described at the Register-Transfer Level (RTL) and how it is verified using testbenches and simulation. In addition, the course provides an overview of the Digital IC Design Flow, including: - Specification and Functional Requirements - RTL Design using HDL - Functional Verification and Testbench Development - Logic Synthesis to Gate-Level Netlist - Timing Analysis and Optimization - Implementation on FPGA or placement & routing (for ASIC) - Post-synthesis simulation and sign-off checks The module emphasizes not only technical competence but also professional ethics, social responsibility, and the development of creativity and problem-solving skills. Students apply theoretical knowledge through laboratory exercises, hands-on projects, and real-world design practices.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU FLM at https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Specific Tools: 1) Simulation software - Cadence - https://www.cadence.com/en_US/home/tools.html - Tinkercad.com 2) Lab equipments/ Training Kits - Tool: Oscilloscope and Function generator, multimeter - Kit for the lab: List of components in the sheet "Electronics component list"
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	968/QĐ-ĐHFPT dated 08/07/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/7/2026

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Digital Design with an Introduction to the Verilog HDL	M. Morris Mano and Michael D. Ciletti	Pearson	2018	6 th	9781292231167	True	True	False	
2	Fundamentals of Digital Logic with VHDL Design	Stephen Brown and Zvonko Vranesic	Mc Graw Hill	2023	4 th	978-1-260-59778-3	False	True	False	
3	Digital Fundamentals	Thomas L. Floyd	Pearson	2015	11 th	978-1-292-07598-3	False	True	False	
4	CMOS VLSI Design: A Circuits and Systems Perspective	Neil H. E. Weste, David Money Harris	Pearson	2011	4th	978-0-321-54774-3	False	True	False	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Explain number systems, conversions, codes, and binary arithmetic.
2	CLO2	Apply Boolean algebra and minimization to design logic circuits.
3	CLO3	Design and analyze combinational logic circuits.
4	CLO4	Model and verify logic circuits using HDL.
5	CLO5	Analyze and design sequential circuits using FSMs.
6	CLO6	Design and implement registers and counters.
7	CLO7	Describe memory and PLDs including FPGA resources.
8	CLO8	Apply RTL design and HDL coding for digital systems.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction Chapter 1: Digital Systems and Binary Numbers 1.1 Digital Systems 1.2 Binary Numbers 1.3 Number - Base Conversions 1.4 Octal and Hexadecimal Numbers 1.5 Complements	Offline	CLO1	IT	Textbook: Digital Design, slides, EduNext	DCD301	Read/review and practice	
2	Chapter 1: Digital Systems and Binary Numbers (cont) 1.6 Signed Binary Numbers 1.7 Binary Codes 1.8 Binary Storage and Registers 1.9 Binary Logic	Offline	CLO1	T	Textbook: Digital Design, slides, EduNext		Read/review and practice	
3	Chapter 1: Digital Systems and Binary Numbers (cont) Exercise: Conversion drills	Offline	CLO1–CLO6	TU	Worksheet, exercises		Solve problems, submit answers	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	Chapter 2: Boolean Algebra and logic Gates 2.1 Introduction 2.2 Basic Definitions 2.3 Axiomatic Definition	Offline	CLO2	T	Textbook, slides, EduNext		Read/review and practice	
5	Chapter 2: Boolean Algebra and logic Gates (cont) 2.4 Theorems 2.5 Boolean Functions 2.6 Canonical and Standard Forms	Offline	CLO2	T	Textbook, slides, EduNext		Read/review and practice	
6	Chapter 2: Boolean Algebra and logic Gates (cont) 2.7 Other Logic Operations 2.8 Digital Logic Gates 2.9 Integrated Circuits	Offline	CLO2	T	Textbook, slides, EduNext		Read/review and practice	
7	Chapter 2: Boolean Algebra and logic Gates (cont) Exercise: Boolean simplification 1	Offline	CLO1–CLO6	TU	Worksheet, exercises		Solve problems, submit answers	
8	Chapter 2: Boolean Algebra and logic Gates (cont) Exercise: Boolean simplification 2	Offline	CLO1–CLO6	TU	Worksheet, exercises		Solve problems, submit answers	
9	Chapter 3: Gate-Level Minimization(cont) 3.7 Other Two-Level Implementations 3.8 XOR 3.9 HDL 3.10 Truth Tables in HDL	Offline	CLO2	T	Textbook, slides, EduNext		Read/review and practice	
10	Chapter 3: Gate-Level Minimization(cont) Exercise: Minimization practice	Offline	CLO1–CLO6	IT	Worksheet, exercises		Solve problems, submit answers	
11	Chapter 3: Gate-Level Minimization(cont) Exercise: HDL practice	Offline	CLO1–CLO6	T	Worksheet, exercises		Solve problems, submit answers	
12	Review (Ch.1–3)	Offline	CLO1–CLO8	T	Notes, textbook, slides		Revise, ask questions	
13	Progress Test 1 (Ch.1–3)	Offline	CLO1–CLO2	TU	Exam paper		Answer under exam conditions	
14	Lab 1.1: Number Systems	Offline	CLO1,CLO2	TU	Lab guide, FPGA board, software		Perform lab tasks, write report	
15	Lab 1.2: Boolean Simplification	Offline	CLO1,CLO2	TU	Lab guide, FPGA board, software		Perform lab tasks, write report	
16	Lab 1.3: K-map Practice	Offline	CLO1,CLO2	TU	Lab guide, FPGA board, software		Perform lab tasks, write report	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
17	Chapter 4: Combinational logic 4.3 Analysis of Combinational Circuits 4.4 Design Procedure 4.5 Binary Adder - Subtractor	Offline	CLO3,CLO4	T	Textbook, slides, EduNext		Read/review and practice	
18	Chapter 4: Combinational logic (cont) 4.6 Decimal Adder 4.7 Binary Multiplier 4.8 Magnitude Comparator	Offline	CLO3,CLO4	T	Textbook, slides, EduNext		Read/review and practice	
19	Chapter 4: Combinational logic (cont) 4.9 Decoders 4.10 Encoders 4.11 Multiplexers	Offline	CLO3,CLO4	T	Textbook, slides, EduNext		Read/review and practice	
20	Chapter 4: Combinational logic (cont) Exercise: Combinational circuits 4.12 HDL Models of Combinational Circuits 4.13 Behavioral Modeling	Offline	CLO1–CLO6	T	Worksheet, exercises Textbook, slides, EduNext		Solve problems, submit answers Read/review and practice	
21	4.14 Writing a Simple Testbench 4.15 Logic Simulation	Offline	CLO3,CLO4	TU	Exam paper Textbook, slides, EduNext		Answer under exam conditions Read/review and practice	
22	Chapter 4: Combinational logic (cont) Exercise: HDL for Combinational Logic	Offline	CLO3,CLO4	TU	Worksheet, exercises		Solve problems, submit answers	
23	Chapter 5: Synchronous Sequential Logic 5.1 Introduction 5.2 Sequential Circuits 5.3 Storage Elements: Latches 5.4 Storage Elements: Flip-Flops	Offline	CLO5	T	Textbook, slides, EduNext		Read/review and practice	
24	Chapter 5: Synchronous Sequential Logic (cont) 5.5 Analysis of Clocked Sequential Circuits 5.6 Synthesizable HDL Models of Sequential Circuits 5.7 State Reduction and Assignment	Offline	CLO5	T	Textbook, slides, EduNext		Read/review and practice	
25	Assignment 1	Offline	CLO1, CLO2, CLO3, CLO4, CLO6	IT	Slides:Instructional Guide for Assignment 1		Assign work to team members Proceed to do the Assignment Answer Edunext questions	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
26	Chapter 5: Synchronous Sequential Logic (cont) 5.8 FSM Design Procedure Exercise: FSM examples Exercise: FSM practice	Offline	CLO1–CLO6	TU	Textbook, slides, EduNext Worksheet, exercises		Read/review and practice Solve problems, submit answers	
27	Chapter 6: Registers and Counters 6.1 Registers 6.2 Shift Registers 6.3 Ripple Counters	Offline	CLO6	T	Textbook, slides, EduNext		Read/review and practice	
28	Chapter 6: Registers and Counters (cont) 6.4 Synchronous Counters 6.5 Other Counters 6.6 HDL Models	Offline	CLO6	T	Textbook, slides, EduNext		Read/review and practice	
29	Chapter 6: Registers and Counters (cont) Exercise: Counters practice	Offline	CLO1–CLO6	T	Worksheet, exercises		Solve problems, submit answers	
30	Chapter 7: Memory and Programmable logic 7.1 Introduction 7.2 RAM 7.3 Memory Decoding 7.4 Error Detection & Correction	Offline	CLO7	T	Textbook, slides, EduNext		Read/review and practice	
31	Chapter 7: Memory and Programmable logic (cont) Exercise: Error Detection & Correction	Offline	CLO7	TU	Worksheet, exercises		Solve problems, submit answers	
32	Chapter 7: Memory and Programmable logic (cont) 7.5 ROM 7.6 PLA 7.7 PAL 7.8 Sequential PLDs	Offline	CLO7	TU	Textbook, slides, EduNext		Read/review and practice	
33	Chapter 8: Design at the Register Transfer Level 8.1 Introduction 8.2 RTL Notation 8.3 RTL Descriptions	Offline	CLO8	TU	Textbook, slides, EduNext		Read/review and practice	
34	Chapter 8: Design at the Register (cont) 8.4 ASMs 8.5 ASMD Chart Example 8.6 HDL Example	Offline	CLO8	TU	Textbook, slides, EduNext		Read/review and practice	
35	Chapter 8: Design at the Register (cont) 8.7 Sequential Binary Multiplier 8.8 Control Logic 8.9 HDL Description	Offline	CLO8	TU	Textbook, slides, EduNext		Read/review and practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
36	Chapter 8: Design at the Register (cont) 8.10 Design with Multiplexers 8.11 Race-Free Design 8.12 Latch-Free Design 8.13 System Verilog Intro	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
37	Chapter 8: Design at the Register (cont) Exercise: RTL Modeling and ASM-Based Control Design in Digital Systems	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
38	Chapter 8: Design at the Register (cont) Exercise: HDL Implementation and Optimization of a Sequential Binary Multiplier	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
39	Chapter 9: Asynchronous Sequential logic 9.1 Introduction 9.2 Analysis Procedure 9.3 Circuits with latches 9.4 Design Procedure	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
40	Chapter 9: Asynchronous Sequential logic (cont) 9.5 Reduction of Stateand FlowTables 9.6 Race - Free StateAssignment 9.7 Hazards	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
41	Chapter 9: Asynchronous Sequential logic (cont) 9.8 Design Example 1	Offline	CLO8	T	Textbook, slides, EduNext		Solve problems, submit answers	
42	Chapter 9: Asynchronous Sequential logic (cont) 9.8 Design Example 2	Offline	CLO8	TU	Textbook, slides, EduNext		Solve problems, submit answers	
43	View Assignment 1	Offline	CLO5, CLO6	U	Slides:Instructional Guide for Assignment 1		Students present their report and demonstrate the product.	
44	Assignment 2	Offline	CLO5, CLO6, CLO7, CLO8	IT	Slides:Instructional Guide for Assignment 2		Assign work to team members Proceed to do the Assignment Answer Edunext questions	
45	Chapter 10: Digital Integrated Circuits 10.1 Introduction 10.2 Special Characteristics 10.3 Bipolar - Transistor Characteristics	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
46	Chapter 10: Digital Integrated Circuits (cont) 10.4 RTI and DTL Circuits 10.5 Transistor-Transistor Logic 10.6 Emitter - Coupled Logic 10.7 Metal - Oxide Semiconductor	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
47	Chapter 10: Digital Integrated Circuits (cont) 10.8 Complementary MOS 10.9 CMOS Transmission Gate Circuits 10.10 Switch - Level Modeling with HDL	Offline	CLO8	T	Textbook, slides, EduNext		Read/review and practice	
48	Chapter 10: Digital Integrated Circuits (cont) Exercise: Mixed - Logic Design and Analysis Using RTL, TTL, and CMOS Technologies	Offline	CLO8	T	Textbook, slides, EduNext		Solve problems, submit answers	
49	View Assignment 2	Offline	CLO5, CLO6, CLO7, CLO8	U	Slides:Instructional Guide for Assignment 2		Students present their report and demonstrate the product.	
50	Lab 2 Lab 2.1: Combinational Circuits on FPGA	Offline	CLO3, CLO5, CLO7, CLO8	TU	Lab guide, FPGA board, software		Perform lab tasks, write report	
51	Lab 2.2: Sequential Circuits on FPGA	Offline	CLO3, CLO5, CLO7, CLO8	TU	Lab guide, FPGA board, software		Perform lab tasks, write report	
52	Lab 2.3: RTL Design Project	Offline	CLO3, CLO5, CLO7, CLO8	TU	Lab guide, FPGA board, software		Perform lab tasks, write report	
53	Lab 3: CMOS AND Gate design lab 3.1 Set up the Cadence environment	Offline	CLO5, CLO6, CLO7, CLO8	TU	Lab guide, Cedance software		Perform lab tasks, write report	
54	Lab 3: CMOS AND Gate design lab 3.2 Part A: Schematic design (Transistor-Level)	Offline	CLO5, CLO6, CLO7, CLO8	TU	Lab guide, Cedance software		Perform lab tasks, write report	
55	Lab 3: CMOS AND Gate design lab 3.3 Part B: Simulation (ADE / Specter)	Offline	CLO5, CLO6, CLO7, CLO8	TU	Lab guide, Cedance software		Perform lab tasks, write report	
56	Lab 3: CMOS AND Gate design lab 3.4 Part C: Layout Design (Layout Editor)	Offline	CLO5, CLO6, CLO7, CLO8	TU	Lab guide, Cedance software		Perform lab tasks, write report	
57	Lab 3: CMOS AND Gate design lab 3.5 Part D: LVS & Extraction (Layout vs Schematic & Parasitic Extraction)	Offline	CLO5, CLO6, CLO7, CLO8	TU	Lab guide, Cedance software		Perform lab tasks, write report	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
58	Review	Offline	CLO3–CLO6	U	Notes, textbook, slides		Revise, ask questions	
59	Progress Test 2 (Ch.4–8)	Offline	CLO3–CLO6	U	Exam paper		Answer under exam conditions	
60	Final Review & Wrap-up	Offline	CLO1–CLO8	U	Notes, textbook, slides		Revise, ask questions	

0 Constructive question(s)

8 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignments 1	on-going	1	5.0%	>0	At home	CLO1 - CLO4	Homework, mini-projects, and optionally the constructivist approach as proposed by the lecturer.	by instructor	Chapters 1–4	Individual or team work, guided by instructor, submission by a given deadline	
2	Assignments 2	on-going	1	5.0%	>0	At home	CLO5 - CLO8	Homework, mini-projects, and optionally the constructivist approach as proposed by the lecturer.	by instructor	Chapters 5–8	Individual or team work, guided by instructor, submission by a given deadline	
3	Lab1	on-going	1	10.0%	>0	135'	CLO1 - CLO4	Electronic Circuit Design, Analysis and Implementation	by instructor	Ch.1–3 (number systems, Boolean, K-map);	Supervised by instructor	
4	Lab2	on-going	1	10.0%	>0	135'	CLO3, CLO5, CLO8	Electronic Circuit Design, Analysis and Implementation	by instructor	Ch.4–6 (combinational & sequential design, RTL)	Supervised by instructor	Individual or small group
5	Lab3	on-going	1	10.0%	>0	225'	CLO6, CLO7, CLO8	Electronic Circuit Design, Analysis and Implementation	by instructor	ch.8,9,10	Individual or team work, guided by instructor, report submission	Hardware-based + simulation
6	Progress Tests 1	on-going	1	10.0%	>0	30'	CLO1–CLO3	Multiple choice + short answer	20	Chapters 1–3	in class, by instructor	Conducted in class
7	Progress Tests 2	on-going	1	10.0%	>0	30'	CLO4–CLO6	Multiple choice + short answer	20	Chapters 4–6	in class, by instructor	Conducted in class
8	Final Exam	Final	1	40.0%	4	60′	All CLOs	Multiple choices Marked by Computer	50	All chapters (Ch.1–10); both theoretical and practical aspects	By Exam board	Closed book, supervised